



sokonalysis

The Cipher Toolkit Built For All Skill Levels

User Guide

Table of Contents

Introduction.....	3
Requirements.....	3
Linux.....	3
Windows.....	3
Features.....	4
Memory/Cashe.....	4
Header.....	4
File.....	4
Exit.....	4
View.....	5
Switch Theme.....	5
Layout.....	5
Default.....	6
Compact.....	6
Wide.....	6
Tools.....	6
Config.....	7
Wordlist Settings.....	7
John Settings.....	11
Quadgram Settings.....	11

Introduction

Welcome to sokonalysis v4.0.0 a cryptographic desktop application meant for all skill levels that can be accessed in both Windows and Linux. This application was developed by Soko James and Rosaria Phiri from Kapasa Makasa University featuring a Command Line Interface (CLI) as their Cybersecurity final year project.

Requirements

Linux

The current version works well on all distros including Kali, Parrot and Ubuntu. Ensure to have the following tools installed externally -

1. John The Ripper
2. Aircrack-ng Suite
3. Steghide
4. Stegseek
5. OpenSSL

Windows

The current version works well in Windows 10 and 11. For cracking hashes, John The Ripper as been integrated in sokonalysis and can be used without any errors. For other options requiring external tools can not be utilized in Windows for now. We recommend, you use Linux or wait for coming updates that will integrate those external tools.

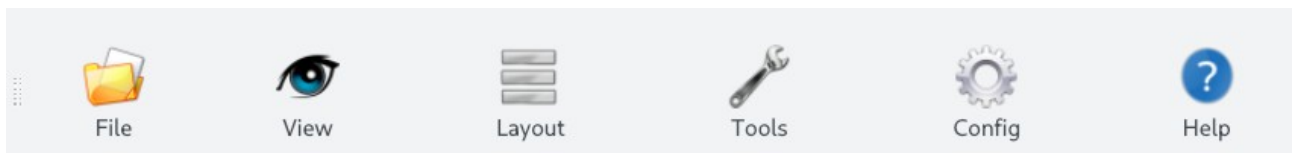
Features

Memory/Cashe

All your preferences such as theme, layout, header position, wordlist and quadgrams paths will be cashed. Your settings will restart where you ended even if you close the app.

Header

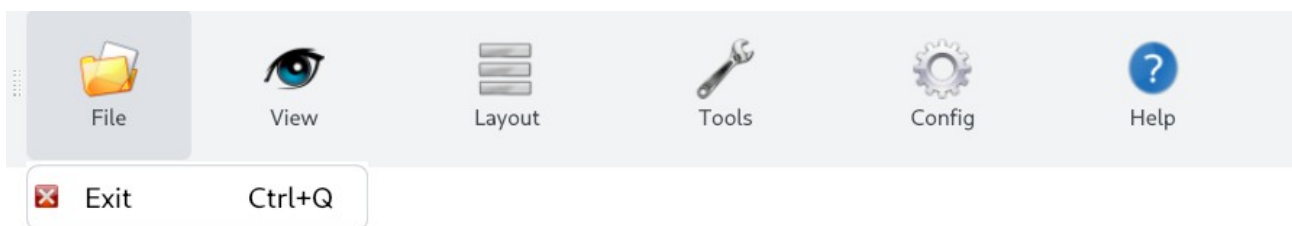
The header is customizable from top to left or right depending on your preference. To achieve this, drag  from the left side of the header to any position of your choice, e.g., left, right or top again.



File

Exit

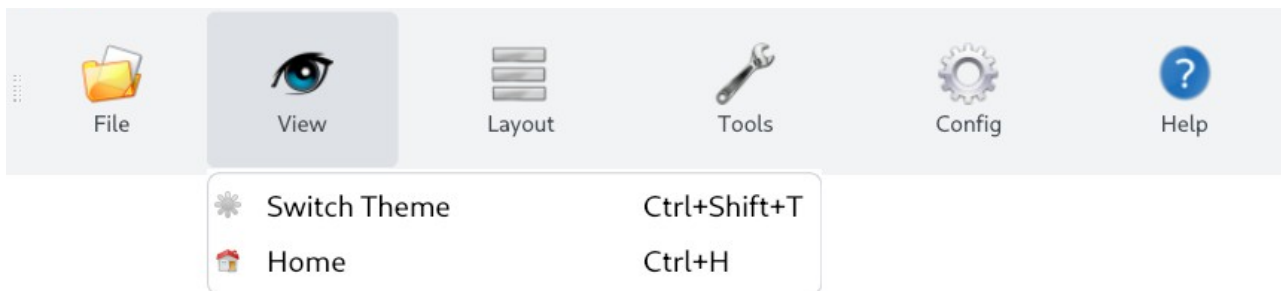
This enables the user to **close** the application.



View

Switch Theme

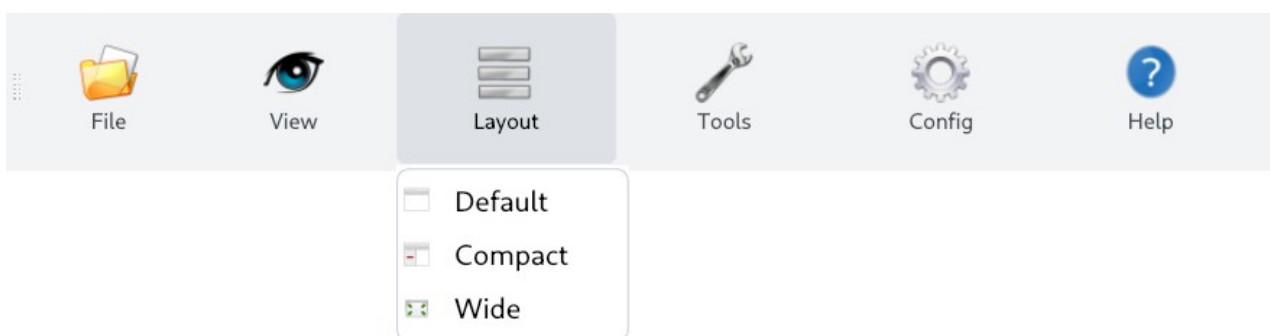
By **default** the applications theme is **light** mode when the applicaion is installed, but will depend on your preference later.



To switch into **dark mode**, you can click on Switch Theme under View or use the shortcut, **Ctrl + Shift + T** on the keyboard or tell the Assistance to help you achieve this.

Layout

When you maximize the window, it will trigger Wide mode, while when minimized it will trigger compact mode.



Default

By default the application opens in the Default layout when you just install the app, then later depends on your preference. This opens the application in a **minimized screen** window mode.

Compact

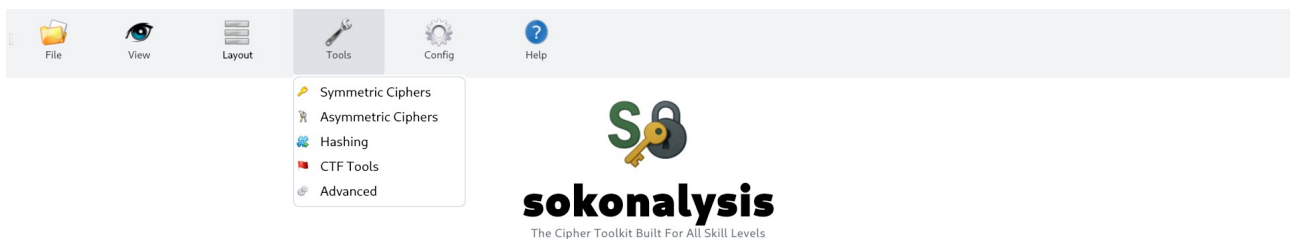
This layout opens the application in a smaller **minimized screen** window mode than the Default layout.

Wide

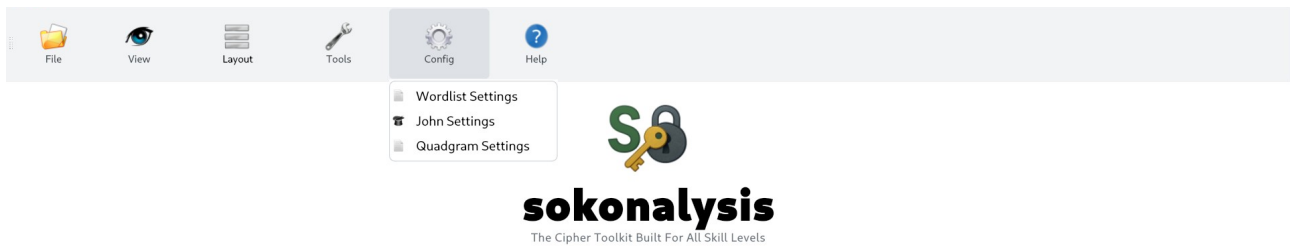
This layout opens the application in **full screen**.

Tools

These are just **shortcuts** of the **main menu**, and can be used when a user is in a category and wants to switch into another category without navigating back multiple times.



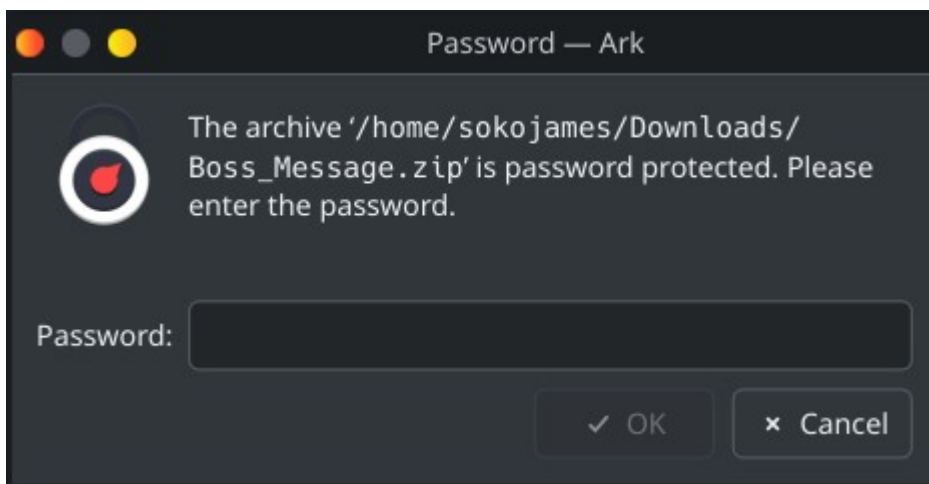
Config



Wordlist Settings

Based on the wordlist used, the system will **slit** it into **multiple parts** based on the system cores or threads, users can also set up the number of parts to be slit. This helps speed up cracking process.

For example, you want to crack a protected documents password, in this case for a CTF challenge on Cyber Talents called Boss Message.



If you are just getting started in CTFs, you can just **darg and drop** the **file** onto the **Assistant**, to get some info.

I see you dropped **Boss_Message.zip** (.zip file)!

This can be processed using **Documents > Zip File**.

Ask Assistant about any tool... (or drop files here)

Now we know that we need to use a wordlist and the **path** for the Zip File cracking as been provided.

You'll need a **wordlist** configured. [Click here to open Wordlist Settings](#) or type '**open wordlist**' to configure now.

[Click here to open Zip File](#)

Ask Assistant about any tool... (or drop files here)

Notice we need to **configure** the **wordlist** first

[Back](#) **Crack ZIP File Password**

 **No Wordlist**  **John Ready**

Using the Config tab then Wordlist Settings can help us achieve this. In most cases, the mostly commonly used wordlist is **rockyou.txt** which can be found via **/usr/share/wordlists/** in a Linux system file structure.

The screenshot shows the 'Configurations' page in the Sokonalysis application. The top navigation bar includes icons for File, View, Layout, Tools, Config (selected), and Help. The main heading is 'Configurations' with a 'Back' link and a 'Wordlist Ready' status indicator. Below the heading, it says 'Configure wordlist settings for all hashing and cracking operations.'

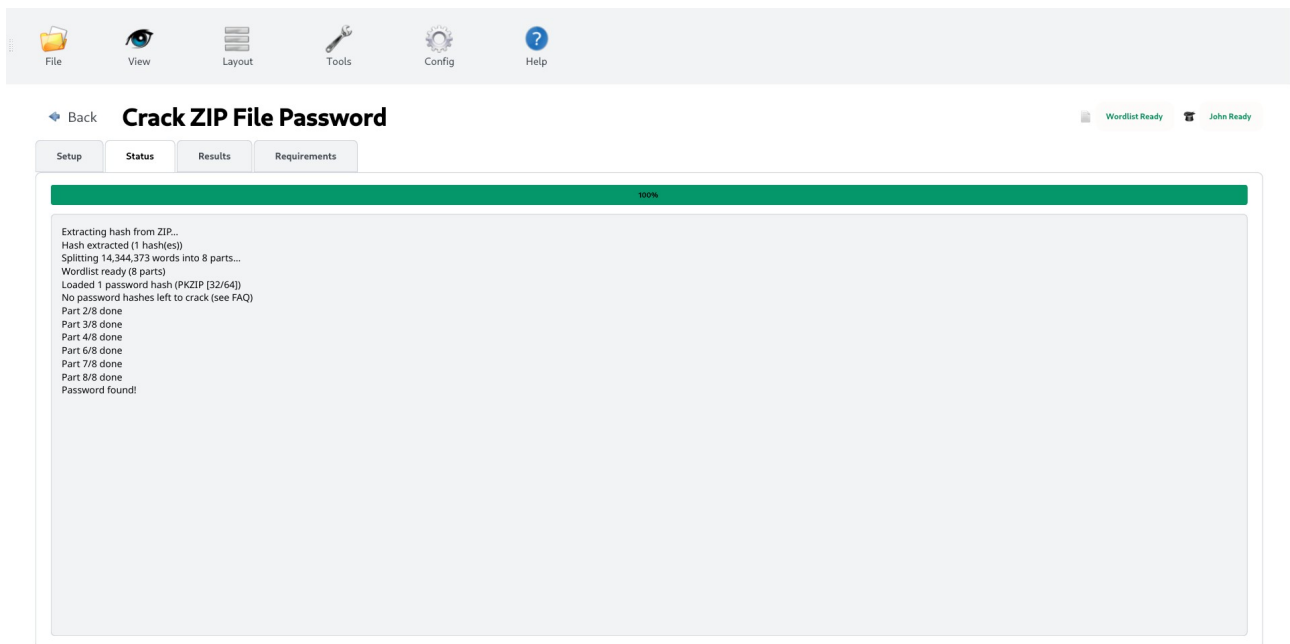
The 'Wordlist File' section contains a dashed box representing the wordlist file. Inside the box, 'rockyou.txt' is listed with a size of '133.4 MB'. Below the box, the text '~17,490,187 words • 133.4MB' is displayed. A text input field shows the path '/home/sokojames/Desktop/Cyber Security/Presentation/rockyou.txt' with a 'Browse' button to its right.

The 'Wordlist Splitting' section has a checkbox labeled 'Auto-detect optimal split count' which is checked. Below this, a 'Number of parts:' label is followed by a text input field containing the value '8'. A small note below the input field states 'Recommended: 8 parts (based on ~17,490,187 words, 8 CPU cores)'.

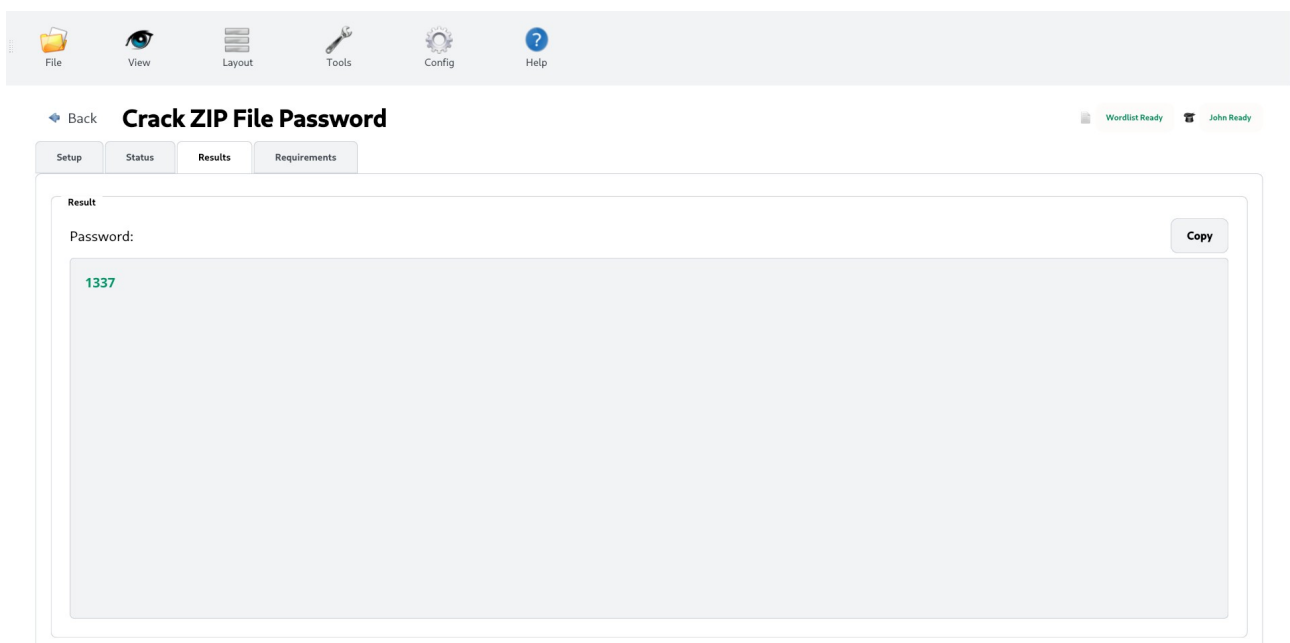
At the bottom of the configuration area is a 'Save Configuration' button.

Save the **configurations**, then drag and drop your zip file then press **Start Cracking**.

Notice the **wordlist splitting** taking place and the password was found in less than a minute.

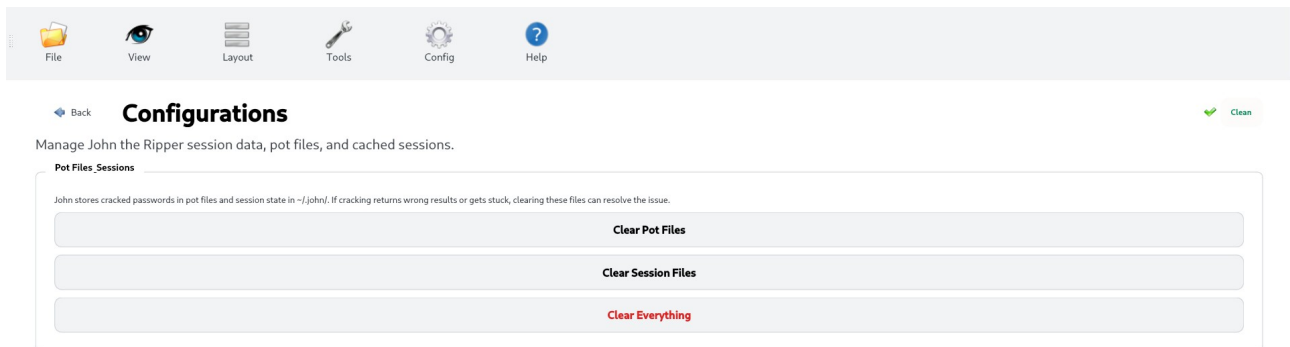


We can use the found password to unzip the Zipped File.



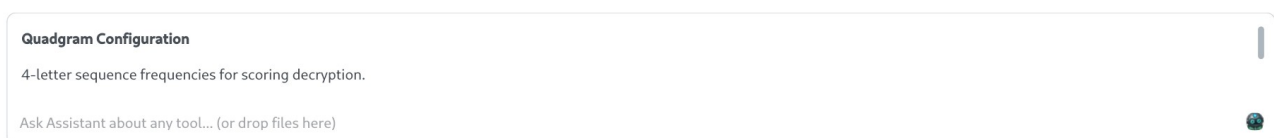
John Settings

Most of the Document password cracking passwords such as Zip File use **John The Ripper**, sometimes john **crashes** its **pot** files where the cracked passwords are saved. So in cases where the john fails to crack a password using a wordlist but the password is in the wordlist, clear the session logs via **Config** the **John Settings**.

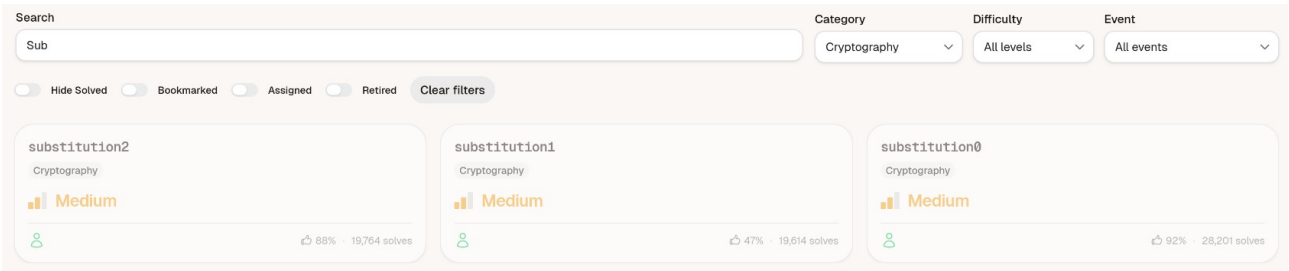


Quadgram Settings

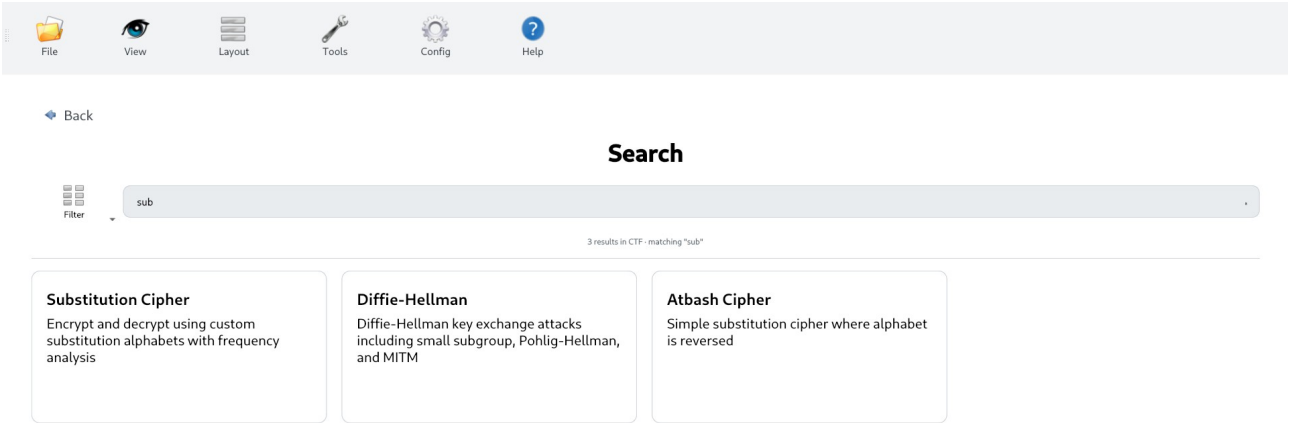
This is useful in a **Substitution Cipher** brute-force cracking using **Frequency Analysis**. The AI assistant can provide us with more info about it.



Let's take an example, on **CyLab Security Academy** famously known as **PicoCTF**, they are multiple **substitution cipher challenges** we can work with.



In this demo, will use **substitution2** challenge. Download the message.txt file to view the cipher text. Using the **Search** option, type sub and **Filter** by the **CTF** category.



[illegible]

File

View

Layout

Tools

Config

Help

Back

Substitution Cipher - Frequency Analysis

Default (EN,json)

Default

Hill-Climbing Settings

Max Rounds:

Consolidate:

10000

3

Input

Status

Results

Decryption Results

Recovered Key:

Alphabet: abcdefghijklmnopqrstuvwxyz xhkyienrgowugmvidfapctbz

Decrypted Text:

ngcnfsgiscriptsoffenseontheotherhandisheavilyfocusedonexplorationandimprovisationandoftenhaselmentsofplaywebelieveacompetitiontouchingontheoffensiveeleme
ntsofcomputersecurityisthereforeabettervehiclefortechevangelismtostudentsinamericanhighschoolsurtherwebelievethatanunderstandingoffensivetechniquesissen
tiaformountinganeffectivedefenseandthatthetoolsandconfigurationfocusencounteredindefensivecompetitionsdoesnotleadstudentstoknowtheirenemyaseffectivelyaste
achingthemtoactivelythinklikeanattackerpicotcfisanooffensivelyorientedhighschoolcomputersecuritycompetitionthatseekstogenerateinterestincomputerscienceamonghi
ghschoolersteachingthemenoughaboutcomputersecuritytopiquetheircuriositymotivatingthemtoexploreontheirownandenablingthemtobetterdefendtheirmachinesthefla
gis

Similary to the Wordlist Settings, we can also **configure** a **custom** quadgram json file. First, switch from **Default (EN.json)** to **Custom Json**.

The screenshot shows the 'Substitution Cipher - Frequency Analysis' tool. At the top, there's a navigation bar with icons for File, View, Layout, Tools, Config, and Help. Below the title bar, there's a 'Back' button and a dropdown menu set to 'Custom JSON' with a red 'No JSON' button next to it. The main section is titled 'Hill-Climbing Settings' and contains two input fields: 'Max Rounds' set to '10000' and 'Consolidate' set to '3'. Below this are three tabs: 'Input', 'Status', and 'Results'. The 'Results' tab is active, showing 'Decryption Results'. It includes a 'Recovered Key' section with a text box containing 'Alphabet: abcdefghijklmnopqrstuvwxyz Key: xhvjlenqrowgmidsfapctzb'. Below that is a 'Decrypted Text' section with a large text area containing a paragraph of text, with the last line highlighted in blue: 'gispicoCTF{N6R4M_4N41Y515_15_73D10U5_42EA1770}'.

Under **Config** select **Quadgram Settings** then **drag and drop** your custom **json** file.

The screenshot shows the 'Configurations' page. At the top, there's a navigation bar with icons for File, View, Layout, Tools, Config, and Help. Below the title bar, there's a 'Back' button and a dropdown menu set to 'Custom JSON' with a red 'No JSON' button next to it. The main section is titled 'Configurations' and contains a subtitle 'Configure the quadgram JSON file used for substitution cipher frequency analysis.' Below this is a 'Quadgram JSON File' section with a large dashed box for dragging and dropping a file. Below the box is a text box saying 'No quadgram file selected...' and a 'Browse' button. At the bottom, there's a 'Save Configuration' button.